Campus Safety & Incident Analytics System

Database Implementation & System Integration Report

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1. Brief Introduction

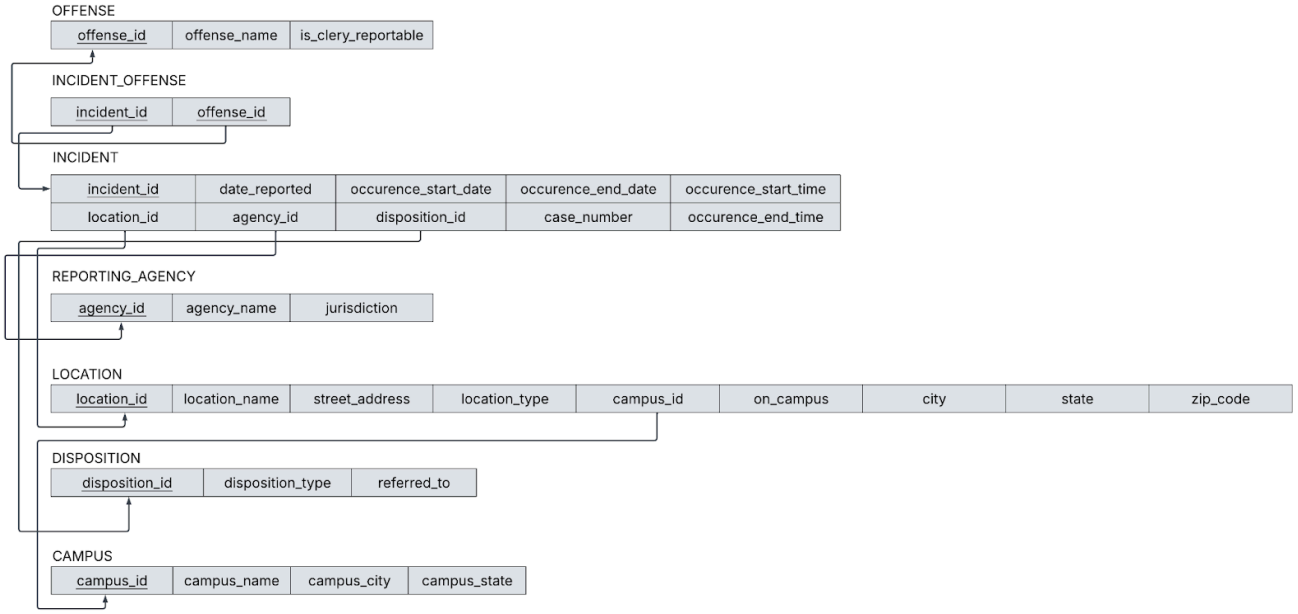
The Campus Safety & Incident Analytics System is designed to store, manage, and analyze campus crime data for Virginia Tech. The system enables users to view associated offenses, locations, reported incidents, and reporting agencies through a structured and normalized relational database.

The application supports querying and filtering incident data, allowing users to retrieve information such as location, dates, offense types, and case numbers through an interactive web interface. By integrating a relational database with a dynamic web-based interface, the system demonstrates how real-world crime data can be efficiently stored, analyzed, and accessed for reporting and decision-making.

The database is populated using structured data extracted from official campus crime reports, ensuring that the system reflects real-world data and supports meaningful analysis.

2. Normalized Relational Schema

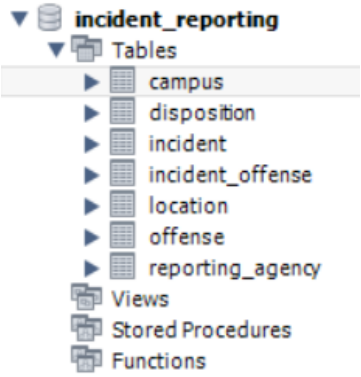
The database schema is normalized to Third Normal Form (3NF), ensuring elimination of redundancy and maintaining data integrity through proper use of primary and foreign keys. Each entity is represented as a separate table, with relationships enforced through foreign key constraints.



3. Screenshots of Tables with Sample Data

All data displayed in the following tables is sourced from official Virginia Tech Police Department crime logs (January 2026). No fabricated data was used. Additional campus and reporting agency information reflects real institutional data.

Incident:



Campus:

campus_id	campus_name	campus_city	campus_state
1	Virginia Tech	Blacksburg	VA
2	Virginia Commonwealth University	Richmond	VA
3	James Madison University	Harrisonburg	VA
4	University of Virginia	Charlottesville	VA
5	George Mason University	Fairfax	VA

Reporting Agency:

agency_id	agency_name	jurisdiction
1	Virginia Tech Police Department	Virginia Tech Campus
2	Virginia Commonwealth University Police Depart...	Virginia Commonwealth University Campus
3	James Madison University Police Department	James Madison University Campus
4	University of Virginia Police Division	University of Virginia Campus
5	George Mason University Police	George Mason University Campus

Location:

location_id	location_name	street_address	location_type	on_campus	campus_id	city	state	zip_code
1	West Ambler Johnston Hall	720 Washington Street SW	NULL	1	1	Blacksburg	VA	24061
2	Public Safety Building	330 Sterrett Drive	NULL	1	1	Blacksburg	VA	24061
3	Cochrane Hall	770 Washington Street	NULL	1	1	Blacksburg	VA	24061
4	Main Eggleston Hall	440 Drillfield Drive	NULL	1	1	Blacksburg	VA	24061
5	Dietrick Hall	285 Ag Quad Lane	NULL	1	1	Blacksburg	VA	24061
6	Outside the Burg Restaurant	204 Draper Road NW	NULL	0	NULL	Blacksburg	VA	24060
7	Surge Space Building	400 Stanger Street	NULL	1	1	Blacksburg	VA	24061
8	O'Shaughnessy Hall	530 Washington Street SW	NULL	1	1	Blacksburg	VA	24061
9	Pritchard Hall	630 Washington Street SW	NULL	1	1	Blacksburg	VA	24061
10	Hoge Hall	570 Washington Street SW	NULL	1	1	Blacksburg	VA	24061
11	Cookout	1311 South Main Street	NULL	0	NULL	Blacksburg	VA	24060
12	Graduate Life Center Lot	170 Otey Street NW	NULL	1	1	Blacksburg	VA	24061
13	Engle Lot	240 West Campus Drive	NULL	1	1	Blacksburg	VA	24061
14	Harper Hall	240 West Campus Drive	NULL	1	1	Blacksburg	VA	24061
15	Goodwin Hall	635 Prices Fork Road	NULL	1	1	Blacksburg	VA	24061
16	New Classroom Building	1455 Perry Street	NULL	1	1	Blacksburg	VA	24061
17	Henderson Hall	195 Alumni Mall	NULL	1	1	Blacksburg	VA	24061
18	Vawter Hall	180 Kent Street	NULL	1	1	Blacksburg	VA	24061
19	East Ambler Johnston Hall	700 Washington Street SW	NULL	1	1	Blacksburg	VA	24061
20	Duck Pond	NULL	NULL	1	1	Blacksburg	VA	24061
21	McComas Hall	895 Washington Street SW	NULL	1	1	Blacksburg	VA	24061

Offense:

offense_id	offense_name	is_clery_reportable
1	Identity Fraud	0
2	Destruction of Property	0
3	Profane, Threatening Language Over Public Air...	0
4	Underage Possession of Alcohol	1
5	Persons Under 21 Driving After Consumption of ...	0
6	Fake ID	0
7	Public Intoxication	0
8	Trespassing	0
9	Obstructing Justice	0
10	Driving Under the Influence of Alcohol	0
11	Aggravated Assault	1
12	Burglary	1
13	Petit Larceny	0
14	Hit and Run	0
15	Sex Offense - Rape	1
16	Sexual Battery	0
17	Stalking	1
18	Concealment, Price Alter Merchandise	0
19	Abduction and Kidnapping	0
20	Assault and Battery - Simple	0

Disposition:

disposition_id	disposition_type	referred_to
1	Active	NULL
2	Inactive	NULL
3	Cleared by Arrest	NULL
4	Inactive	Student Conduct
5	Cleared by Arrest	Student Conduct
6	Inactive	Title IX

Incident:

incident_id	case_number	date_reported	occurence_start_date	occurence_end_date	occurence_start_time	occurence_end_time	location_id	agency_id	disposition_id
1	2026-25	2026-01-02	2026-01-02	NULL	01:00:00	06:05:00	11	1	3
2	2026-98	2026-01-05	2025-12-19	2026-01-05	00:00:00	09:31:00	1	1	2
3	2026-325	2026-01-10	2026-01-03	NULL	00:45:00	00:05:00	2	1	2
4	2026-805	2026-01-17	2026-01-17	NULL	23:56:00	NULL	6	1	5
5	2026-959	2026-01-20	2026-01-20	NULL	09:37:00	NULL	7	1	3
6	2026-1013	2026-01-21	2026-01-21	NULL	01:57:00	02:16:00	7	1	5
7	2026-1060	2026-01-21	2025-11-05	NULL	12:00:00	14:00:00	5	1	4
8	2026-1037	2026-01-21	2025-12-16	2026-01-18	13:35:00	14:11:00	8	1	2
9	2026-1230	2026-01-23	2026-01-22	NULL	00:30:00	01:30:00	9	1	1
10	2026-1746	2026-01-31	2026-01-31	NULL	22:31:00	NULL	5	1	1

Incident_offense:

incident_id	offense_id
1	1
2	2
3	3
4	4
4	6
4	7
5	8
5	9
6	10
7	11
8	12
9	15
9	16
10	17
9	19

4. Confirmation of DBMS-Interface Connection

DBMS Information:

- **Database Management System:** MySQL
- **Backend Interface:** Python (Flask)
- **Frontend Interface:** HTML/CSS with Jinja templates
- **Development Tool:** MySQL Workbench (used for database management)

The system successfully connects the Flask backend to the MySQL database using a database connector. This connection enables the web application to dynamically query, insert, and display incident data through the user interface.

The backend establishes a connection to the database using configured connection parameters (host, user, password, and database name). SQL queries are executed within the Flask application to retrieve and manipulate data, and the results are passed to the frontend using Jinja templates for rendering.

This integration allows for real-time interaction between the user interface and the database. Users can view updated incident records, apply filters, and access analytics without directly interacting with the database. Administrative users have the ability to insert, update, and delete records through controlled application routes, ensuring that all database modifications occur through the backend rather than direct database access.

Additionally, the system enforces data integrity by utilizing structured queries, foreign key relationships, and controlled data insertion logic. Error handling is implemented within the backend to manage failed queries or invalid inputs, preventing inconsistent or incomplete data from being stored.

This confirms that the database and application layers are fully integrated and functioning as intended.